Exp No: 30

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RTOS TASK SCHEDULING

AIM:

To simulate Round-Robin scheduling.

SOFTWARE REQUIRED:

• Ubuntu (or WSL Ubuntu)

• g++

• CMake

• SystemC

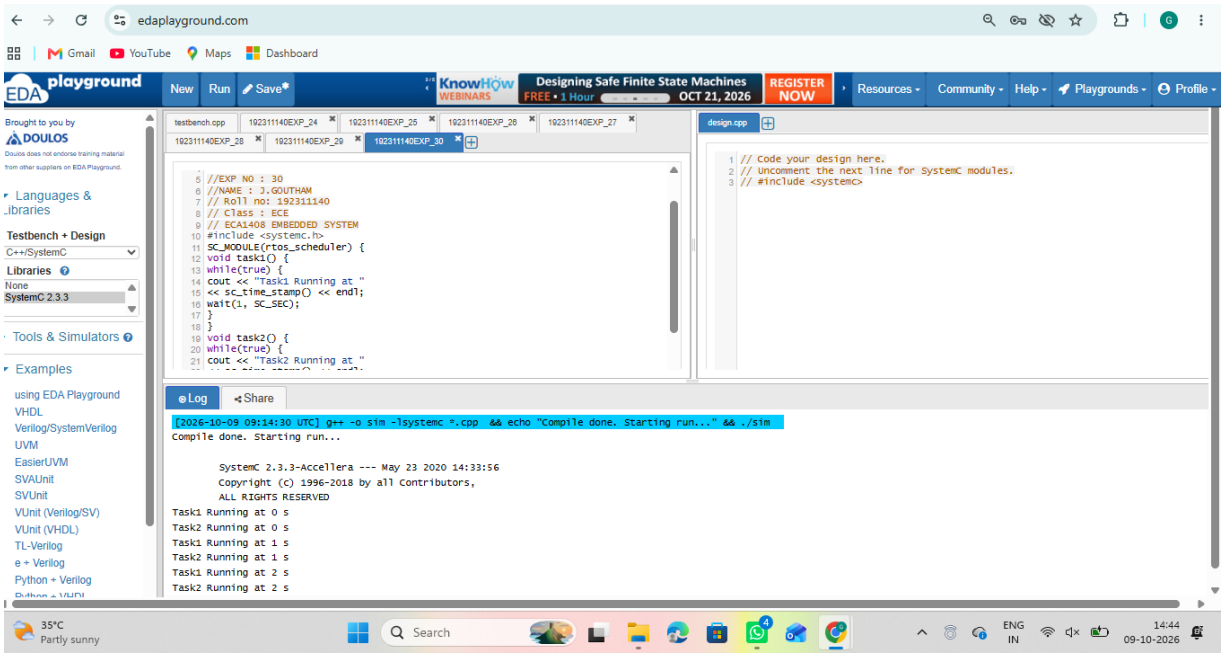
PROGRAM:

```
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() { while(true) {
cout << "Task1 Running at " <<
sc_time_stamp() << endl; wait(1,
SC_SEC);
}
}
void task2() { while(true)
{
cout << "Task2 Running at " <<
sc_time_stamp() << endl; wait(1,
SC_SEC);
```

```
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};

int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
sc_start(6, SC_SEC); return 0;
}
```

OUTPUT:



RESULT:

Thus the Round-Robin scheduling was executed using SystemC and verified.